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CAPITAL MARKET FINANCIAL AND RISK MANAGEMENT.

A market in a general terms represent the set of facilities that make it possible for the exchange of goods and services.

Financial market: This is a specialized form of market in that only securities are traded on a regular basis. Because of the specialized form, the financial market plays an important role in facilitating the smooth running of the economic system.

Types of financial market:

- A The money market and
- B The capital market

Money market: This market deals with short term securities. The forerunners of the market in Nigeria is the West African Currency Board. The board was formed to deal with two main issues.

- 1 To provide the means for financing the export trade needs of the expatriate firms in West Africa and
- 2 To eradicate the confused, inconvenient and unsatisfactory currencies in the British West Africa.

Capital market

The capital market provides financial market in which long term capital is bought and sold or lent and borrowed. The reasons for the establishment of the Nigerian capital market includes:

2) a to provide local opportunities for lending and borrowing of long term purpose.

5) to enable the authorities to mobilize long-term capital

6) to provide foreign businesses the facilities to offer their shares and the Nigerian public the opportunity to invest and participate in the ownership of such foreign businesses.

7) to provide facilities for quotation and ready marketability of shares and stocks and provide opportunities and facilities to raise fresh capital in the market.

8) to provide a code of conduct, check abuses and regulate the activities of the operators in the market

9) to provide a healthy and mutually acceptable environment for participation and cooperation of indigenous and expatriate capital in the joint effort to develop the Nigerian economy.

Membership of the Capital market.

The Nigeria Capital market comprises the following institutions.

1 The Nigerian Stock Exchange

2 The Nigerian Industrial Development Bank limited

3 The Nigerian Bank for Commerce and Industry

4 The Nigerian Agricultural and Rural Development Bank

5 The Federal Mortgage Institution

6 The Securities and Exchange Commission

7 The Insurance Companies

⑧ The National Provident Fund (Social Security Insurance)

⑨ Any organization that has fund for long term investment

It must be noted that the CBN has a supervisory role to play in the capital market too.

Instructions traded in the market
The instrument of the Nigerian capital market includes:

- ① Government development loan stock
- ② Ordinary Shares
- ③ Preference Shares
- ④ Long term debts
- ⑤ Warrants and
- ⑥ Convertible Securities.

Institutions in the capital market or regulators of the financial market

① The Securities and Exchange Commission (SEC)

This is an apex regulating institution of the Nigerian capital market. It is a federal government agency established by the Securities and Exchange Commission Act 71 of 1979, which was re-enacted as Decree number 29 of 1988.

Since its inception, the SEC has been playing within the capital market a similar role to that played by the CBN in the money market.

As a statutory corporation under the supervision of the federal ministry of finance, the Commission submit report of its activities annually to the ministry.

- 4) The Commission, functions, role and structure are set up to achieve the following basic objectives
- ③ Objectives of the Commission
 - ③ Investor protection
 - ⑤ Capital market development towards enhancing socio-economic growth and development
 - ③ Orderly and equitable dealings in securities business.

Structures of SEC.

For the purpose of carrying out its activities the Commission consists of:

- ⑨ A chairman - who is the chairman of the Board of Directors appointed by the President and Commander in Chief of the Armed Forces on the recommendation of the Minister of Finance.
 - ⑤ One person not below the rank of Director to represent the Ministry of Finance.
 - ③ One person not below the rank of Director to represent the CBO
 - ④ The Director General of the Commission
 - ③ Two full time Commissioners who must be persons with ability, experience and special knowledge in Capital market matters
 - ⑤ Five other Commissioners who must be persons with proven ability and expertise in corporate matters generally.
- Apart from the Director-General and the two full time Commissioners, the other members of the Commission are part time members.

5) Functions of the Commission.

④ Regulating: Regulating investment and securities in Nigeria as defined by the Act. This amount to protecting of the investing public from deceit and other unscrupulous practices in the sales of securities which can destabilize the market.

⑤ Registration: This includes the registration and regulations of securities exchange, capital trade points, futures, option and derivatives exchange and any other recognised investment. Exchange and registration of securities to be offered for subscription or sales to the public.

⑥ Investigation. As empowered by the Act, the Commission investigate all reports of violation and suspected violation of security law.

⑦ Enforcement: The SEC has the responsibility of ensuring that participants in the market comply with security laws. It has the power to initiate penalties and remedial actions if there has been any violation.

⑧ Rules making: The Commission makes the rules, create and review procedures as the occasion demands.

⑨ Timing of issues: The SEC is also empowered to control the timing and amount of issue in order to avoid demand on the primary market ~~be~~ beyond its capacity to absorb.

⑩ Review of account: The Commission carries out an regular 5 years review of account of companies covered by SEC.

⑪ The Nigerian Stock Exchange (NSE)

The NSE mission is promoting the Nigerian capital market to respond to the socio-economic development need of the nation.

⑥ The primary aim is to mobilize long term funds for investment.

Function of the MSE.

- ① Provide additional channel for engaging and mobilizing domestic savings for productive investment and represent an alternative investment financing of consumption loans.
- ② Providing depositors with better protection against inflation and currency depreciation
- ③ Fostering the growth of the domestic financial services sector and the various forms of institutional savings such as life insurance and pension funds.
- ④ Improving the gearing of the domestic corporate sector and help reduce dependence on borrowing
- ⑤ Facilitate the transfer of enterprises from the public sector to the private sector.
- ⑥ Provide access to finance for new and smaller companies and encourage institutional development in facilitating the setting up of Nigerian domestic funds, foreign funds and venture capital fund.
- ⑦ Stimulating industrial as well as economic growth and development of the Nigerian economy.
- ⑧ Issuing house.
Is a financial institution that engages in finding private firms wishing to convert to public companies by issuing shares on their behalf

④ Raising of Equity Capital.

The methods of raising equity funds or putting a company in a position for raising equity capital in future, which are particularly to quoted or listed company.

① Stock Exchange Introduction.

This is not a method of raising new capital but of getting permission to deal i.e. introduce the shares of the company to the market. The company after quotation will have access to finance in the capital market in the future.

② Offer for Sale

In this method, the shares are offered to the public including the existing shareholders through. At the end of the offer for sale, the nominal share capital of the company remains unaltered and the proceeds of sales go to the vendor (existing shareholders) and not the company. Offer for sale is not a fresh issue of shares, but sales of existing share by existing shareholders.

③ Offer for Subscription

In this method, the company itself through the agency of an issuing house offers fresh issues to the public. The idea here is to raise supplementary capital for the company. The proceeds of issue go to the company and the number of shares outstanding at end of the exercise will increase.

④ Placing of shares

In this method the shares are offered to a specific group of investors usually insurance companies, pension funds or any other institutional investors.

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(5) Right Issues

In this method, additional finance is obtained from the existing shareholders. This method avoid issuing cost

Leasing.

Lease is a contract between a lessor and a lessee for the hire of a specific asset.

The lessor retains ownership of the asset but conveys the right to the use of the asset to the lessee for an agreed period of time in return for the payment of a specific rent

There are two type of lease

(1) Finance or Capital lease

A Finance lease is a lease that ~~generally~~ ^{normally} lasts for the expected working life of an asset. The lessor receives rental payments from the lessee that covers the full costs of the asset, plus a profit margin. Other characteristics of this contract are.

- * One lease exists for the whole life of the asset
- * The lessor does not usually deal directly in this type of asset
- * The lessor does not ~~receive~~ ^{retain} the risk or reward of ownership
- * The lease agreement usually cannot be cancelled. The lessee has a liability for all payments

* The substance of the transaction is the purchase of the asset by the lessee financed by a loan from the lessor.

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⑤ Operating lease

This is a rental agreement between a lessor and a lessee whereby the lessor is responsible for the upkeep, maintenance, servicing and insurance of the leased asset i.e. all risk and rewards incidental to ownership remain with the lessor.

These type of leased agreement does not cover the useful economic, ~~but~~ working life of the leased asset, so that at the end of an agreement, the lessor can lease the same asset to ~~the~~ someone else (or the same lessee) and obtain a good reward.

Other characteristics include

- * The lessor may vary well carry on trade in this type of asset
- * The lease can sometime be cancelled on short notice. The substance of the transaction is the short term rental of asset.

Advantage of leasing

- ① It provides 100% of the finance of the asset. A bank loan or other form of credit might only provide part of the asset's cost. With leasing a company may preserve liquidity.

- ② The security for the lease is the asset itself. Thus leaves the company's asset (land, building etc) free to be used as security for further debt and might increase the debt raising capacity if a company

- ③ Leases may be easier to arrange than bank loan for small companies without a track record

- ④ Operating lease may still be off the balance sheet and not appear in financial ratios.

(10) Disadvantages of Leasing.

- ① Like any other debt financing, leasing brings some forms of dilution in control.
- ② The rental are payable whether or not profit is made.
- ③ If the rental are fixed and the interest rate in the market falls, the lessee loses.
- ④ In finance lease the risk of ownership is effectively that of the lessee's, should the ~~cost~~ asset becomes obsolete because of high technological changes, he bears any consequential losses.

Portfolio Theory

Portfolio theory is concerned with how investors build a portfolio of investment that gives them suitable balance between return and investment risk.

Risk averse investors will accept risk provided that they are compensated for it by an adequate return.

Diversification to reduce risk.

An investor can reduce the investment risk (in other words, reduce the ~~total~~ volatility of expected returns) by diversifying his investment and holding a portfolio of different investments.

Creating a portfolio of different investments can reduce the variance and standard deviation of returns from the total portfolio, because if some investments provide a lower-than-expected return, others will provide a higher-than-expected return.

Similarly a company could reduce the investment risk in its business by diversifying and building a portfolio of different investments.

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It studies the effect of the diversification of investments on the total risks and returns of a firm. Investment could be negatively correlated or positively correlated.

positively correlated.

When returns from different investments in a portfolio are positively correlated, this means that when ~~one~~ the return from one of the investments is higher than expected, the return from the other investments will also be higher than expected. Similarly, when returns from one investment are lower than expected, the returns from all the investments in the portfolio will be lower than expected.

Negatively correlated.

When return from two different investments in a portfolio are negatively correlated, this means that when the return from one of the investments is higher than expected, the return from the other investment will be lower than expected.

Neither positively or negatively correlated

This means that when the returns from one investment are higher than expected, the returns from the other investment might either be higher or lower than expected. Investment risk is reduced by building up a portfolio of investments whose returns are negatively correlated, or where correlation is low.

(18) Efficient Market Hypothesis

An efficient market ^{is defined} ~~is defined~~ as a situation where information concerning security is widely and cheaply available to investors and where all relevant and ascertainable information is already reflected in security prices.

Forms of efficient market

I Weak form efficiency
This is a situation where the security prices reflected all the past information as reported by the press.

II Semi Strong form of efficiency
This is a situation where the security prices reflect not only past information but all other published information.

III Strong form of efficiency
This is the situation where the security prices reflect not ~~public~~ information but all information that can be acquired by painstaking analysis of the company and the security.

The Capital Asset Pricing model (CAPM)

It matters how much you diversify your investments. Some level of risk will always exist, so investors naturally seek a rate of return that compensate for that risk. The CAPM helps to calculate investment risk and what return on investment investors should expect.

The risk model consideration is made up of two parts

I Unsystematic risk - changes in regulation and effects on a industry, cutting off a new company, higher, project failures

II Systematic risk.

Systematic risk - This is a risk that is unique to individual investments or securities that can be eliminated through diversification.

Systematic risk.

This is a risk that cannot be diversified away because, it is a risk that affects the market as a whole and all investment in the same way.

Risk and return Investment. Interrelated changes in inflation, recession, war etc.

Investors invest in shares and bonds in the expectation of making a return. The return that they want from investment could be less or more as

- * A return as reward for providing funds and keeping these funds invested plus
- * A return for compensate the investor for the risk

As a basic rule, an investor will expect a higher return when the investment risk is higher.

What is investment risk.

Investors in bonds, investors in shares and company all faces ~~the~~ investment risk. In the case of bonds, the risk for the investor are as follows:

- ① The bond issuer may default, and fail to pay the interest on the bonds, or fail to repay the principal at maturity.

- ② There may be change in market rate of interest, including interest yield on bonds. A change in yields will affect the market value of the bonds. If interest rate rises, the market value of bond will fall and bond investors will suffer a loss in the value of his investment

(14) In case of equity shares! The risk for the investors are that

- * the company might go into liquidation or
- * much more significantly, the company profit might fluctuate, and dividends might also rise or fall from one year to another

When a company invests in a new project, there will be investment risk, the risk that actual return from the investment will not be the same as the expected return but could be higher or lower. The investment risk for the companies is similar to the investment risk facing equity investors.

Some type of investment are more risky than the others because of the nature of the industry and markets. For example investments by a supermarket group in building a new supermarket is likely less risky than investment by an IT company in a new type of software. This is because IT business is inherently more risk than the supermarket business. When business risk is higher, returns are less predictable or more volatile, and the expected returns should be higher to compensate for the higher business risk.

Measurement of risk as a deviation or standard deviation of expected return.

High volatility in returns is associated with high investment risk

Risk can be measured statistically, either from an analysis of historical returns achieved in the past or from probability estimates of returns in the future. The measure of risk is a

measure of the volatility of possible returns.

This volatility can be measured as either the Variance or standard deviation of expected return for an investment

- (a) an expected or average annual return, and
 (b) a variance or standard deviation of the returns. (The standard deviation is the square root of the variance)

Formula: Variance.

$$\sum p(r - \bar{r})^2$$

where

p = the probability of a given return on the investment

r = the amount of that return

$\bar{r}$ = the average expected return on the investment. This is the expected value of the return.

Based on historical analysis, it has been established that the probable returns on equity investment A are.

Probability p	Return r	(EV) of return $p(r)$
0.2	2	0.4
0.5	6	3.0
0.3	8	2.4
Mean		<u>5.8</u>

The Variance of the returns is calculated as follows

Probability p	Return r	$r - \bar{r}$	$(r - \bar{r})^2$	$p(r - \bar{r})^2$
0.2	2	-3.8	14.44	2.888
0.5	6	+0.2	0.04	0.20
0.3	8	+2.2	4.84	1.452
Variance (σ^2)				<u>4.54</u>
Standard deviation (σ)				<u>2.09%</u>

Variance and Standard deviation.

The following information relates to two Securities (X and Y) and how they behave under different set of condition, described as State of the world.

State of the world	probability (p)	Return From Shares	
		X (Rx)	Y (Ry)
A	0.3	25%	22%
B	0.5	22%	18%
C	0.2	12%	15%

You are given the following details concerning the possible returns of two stocks A and B

State of Economy	probability of state of Economy	Security A Returns if state occurs.	Security B Returns
Burn	0.6	40%	10%
Recession	0.4	-10%	30%

Calculate the Variance and Standard deviation in each of the stocks.

The calculations above show that the returns of share X and less volatile than the returns on share Y. In other words, share is less risky than share Y

Share X variance 21.49%,
Std dev. (sx) 4.64%

Y variance 26.64%,
Std dev (sy) 5.16%